Wesley Rancher

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University of Oregon
Terrestrial Ecosystems Ecology and Landscapes Lab
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Summary

Environmental data scientist with expertise in remote sensing, GIS, and landscape ecology. Experienced in applying machine learning to large-scale spatial datasets to track ecosystem change, with a focus on carbon dynamics and resilience. I use data-driven insights to support sustainable infrastructure, guide environmental management, and inform decisions that connect natural and built systems.

Education

University of Oregon | M.S. in Geography, GPA: 3.9
Ohio Wesleyan University | B.A. in Environmental Studies and Geography, GPA: 3.5

Skills

- **GIS & Remote Sensing:** ArcGIS Pro, QGIS, Google Earth Engine; experience with LiDAR, multispectral, and drone imagery
- **Programming:** R (terra, tidymodels, randomForest) and Python (geopandas, rioxarray, scikit-learn); Bash, Git, Docker, AWS CLI
- **Modeling & Analysis:** Random Forest, LANDIS-II, and machine learning and simulation tools for ecosystem and climate applications
- **UAS:** FAA Part 107 Remote Pilot (#4802988); worked with DJI Pilot, Pix4D, Drone2Map, Agisoft; LiDAR and dual Red-Edge sensor calibration
- **Languages:** Spanish (working proficiency)

Research Experience

Graduate Research Assistant – University of Oregon — September 2023 – Present
Advisor: Dr. Melissa Lucash

- Modeled and mapped ecosystem dynamics and aboveground biomass using machine learning in R and Python on large geospatial datasets
- Automated Landsat time series preprocessing in Google Earth Engine, including atmospheric/topographic correction and cross-sensor calibration
- Built reproducible, scalable ecological simulations by containerizing LANDIS-II with Docker
- Integrated and managed large climate datasets (CMIP5/6) to improve predictive modeling
- Collaborated with Bonanza Creek LTER to analyze forest change and wildfire impacts in boreal ecosystems

Graduate Research Fellow – NASA DEVELOP — June 2023 – August 2023
Advisor: Dr. Anthony Vorster

- Mapped invasive plant communities in Utah's Paria River Watershed for Grand Staircase Escalante Partners using remote sensing and field data
- Integrated vegetation indices, senescence metrics, and field observations with random forest models to predict plant cover
- Processed LiDAR and Landsat datasets with ArcGIS and Google Earth Engine, automating spatial analyses for large-scale mapping

Undergraduate Research Assistant – Ohio Wesleyan University — December 2022 – May 2023
Advisor: Dr. Nathan Rowley

- Modeled supraglacial lake depth in Western Greenland using radiative transfer models

- Built reproducible Landsat processing workflows for raster sieving and feature detection
- Calibrated dual-red-edge (Micasense) and LiDAR sensors on DJI drones
- Prepared educational material for FAA Part 107 certification

Undergraduate Research Fellow – University of Central Oklahoma June 2022 – July 2022
Advisor: Dr. Victor Gonzalez

- Participated in NSF-funded REU analyzing climate stressors on heat tolerance of honeybees and sweat bees in Lesvos, Greece
- Built testing apparatuses, conducted fieldwork, and contributed to methodology for temperature acclimation, starvation, and thermal limit assays
- Found bees maintain heat tolerance following desiccation and starvation

Teaching Experience

Graduate Teaching Assistant – University of Oregon September 2023 – Spring 2025

- **Geography 485/585: Remote Sensing I** Winter 2025, Fall 2024
  - Developed lab exercises, taught GIS and remote sensing software (ArcGIS, QGIS, R), and led hands-on demonstrations to undergraduates and graduates
- **Geography 199: Global Wildfire** Spring 2024
  - Supported curriculum development, provided supplemental instruction, and assisted with student questions on wildfire concepts
  - ***Guest lecture:*** *"Changing Wildfire in Brazil"* – discussed landscape drivers of a changing fire regime in Brazil
  - ***Guest lecture:*** *"Bees and Wildfire"* – explained the interplay between post-wildfire effects, vegetation, and pollinators
- **Geography 181: Our Digital Earth** Winter 2024, Fall 2023
  - Facilitated labs focused on digital mapping and spatial data; helped students with ArcGIS Online and digital geography concepts

Awards and Honors

Rippey Research Grant ($1000, UO)	2024
NASA Develop Scholarship ($1500, SSAI)	2023
Dean's List (OWU)	Fall '22, Spring '20, '22, '23
Robert E. Shanklin Distinguished Scholar (Geography, OWU)	2023
Phi Sigma Tau (Philosophy, OWU)	2023
Our New Gold Digital Storytelling winner (Spanish, OWU)	2022

Publications

- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). *Estimating aboveground carbon using machine learning and process-based models. (In preparation)*
- Lucash, M., Lamping, J., Nowell, B., Scheller, R., Banerjee, T., Buettner, C., Fawcett, J., Hurteau, M., Parks, S., Rancher, W., Robbins, Z., St. Denis, L., Stasiewicz, A., Urza, A., & Weiss, S. (2025). *Roadmap for the future of extreme wildfire events. (Submitted)*
- Weiss, S., Rancher, W., Hayes, K., Buma, B., & Lucash, M. (2024). *Wildfire dynamics under climate change in interior Alaska. (In preparation*
- Gonzalez, V., Rancher, W., Vigil, R., Garino-Heisey, I., Oyen, K., Tscheulin, T., Petanidou, T., Hranitz, J., & Barthell, J. (2024). *Bees remain heat tolerant after acute exposure to desiccation and starvation. Journal of Experimental Biology*
- Rowley, N., Rancher, W., & Karmosky, C. (2024). *Comparison of multiple methods for supraglacial melt-lake volume estimation in western Greenland during the 2021 summer melt season. Glacies*

Presentations

- Rancher, W. (2025). *Estimating species-level aboveground carbon in interior Alaska using machine learning and process-based models*. University of Oregon, Eugene, OR. *(Master's thesis presentation)*
- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). *Estimating recent shifts in aboveground carbon and species composition in interior Alaska using Landsat imagery and random forests*. *Northwest Scientific Association*, Eugene, OR. *(Poster)*
- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2024). *Assessing vegetation shifts in boreal Alaska by integrating Landsat imagery with spatial modeling. American Geophysical Union*, Washington, DC. *(Poster)*
- Rancher, W., VanArnam, M., Kowalski, A., Anarella, T., & Vorster, A. (2023). *Mapping Russian olive and tamarisk to inform invasive species management along the Paria River, Utah*. *NASA DEVELOP Day*, Washington, DC. *(Virtual talk)*
- Rancher, W., Rowley, N. (2023). *Estimating supraglacial melt lake volume changes in west-central Greenland using multiple remote sensing methods*. *Ohio Wesleyan Spring Symposium*, Delaware, OH. *(Poster)*
- Rancher, W., Vigil, R., Garino-Heisey, I., & Gonzalez, V. (2022). *Effects of desiccation on bees' heat tolerance*. *Ohio Wesleyan Connection Conference*, Delaware, OH. *(Poster)*
- Rancher, W., & Gonzalez, V. (2022). *Effects of desiccation on bees' heat tolerance. IUSSI Sección Andina y del Caribe*, Panama City, Panama. *(Talk)*